Volume Control Transfers, Conditioning Does Not: The Locality of Implicit EM in Neural Networks

Abstract

Log-sum-exp objectives produce responsibility gradients: for an objective over distances or energies, the derivative with respect to each distance coordinate is the posterior responsibility of the corresponding component. This identity explains how standard neural objectives can instantiate generalized EM without explicit latent-variable updates. We show that the identity is local. In distance coordinates, the LSE Hessian is uniformly bounded by 1/2; with private affine component parameters, this gives a parameter-independent curvature bound. When the same site is composed behind a learned dense map, the leading term of the composed cross-entropy Hessian is a conjugated softmax Hessian whose bound scales with the learned spectrum of the map, so the parameter-independent guarantee is forfeited.

Experiments in supervised MNIST networks support a controlled dissociation: volume control transfers, conditioning does not. Volume control at an intermediate EM site eliminates dead units and reduces redundancy across widths, showing that local mixture failures can be repaired inside a supervised model. The optimizer signature of the single-layer implicit-EM model, however, does not survive ordinary supervised composition: SGD becomes learning-rate sensitive again and the advantage of adaptive methods reappears. Mechanism measurements show the same pattern from inside the model. At the standard auxiliary weight, gradient diagnostics indicate that the cross-entropy path dominates the weighted local EM path by roughly 67–95× late in training, with no detectable alignment or opposition between the two directions. Cutting that path with stop-gradient restores learning-rate insensitivity; increasing the EM weight until the EM path dominates restores the same behavior without cutting the graph. Thus implicit EM is local, graded, and structural: volume control repairs failures at EM sites, but EM conditioning survives only when responsibility gradients reach the relevant parameters directly or dominantly, with simplex-compatible transport remaining a theoretical first-order preservation class.

1 Notation and Conventions

This section fixes the sign convention used throughout the paper. We use *energy* and *distance* interchangeably: lower values indicate better matches. For an input x and K components, d_j denotes the energy of component j. Responsibilities are defined by

$$r = \mathrm{softmax}(-d), \qquad r_j = \frac{\exp(-d_j)}{\sum_k \exp(-d_k)}.$$

Thus $r_j \geq 0$ and $\sum_j r_j = 1$. In this convention, the log-sum-exp objective is the negative log marginal

$$L_{\mathrm{LSE}}(d) = -\log \sum_j \exp(-d_j),$$

and differentiating with respect to a distance gives

$$\frac{\partial L_{\text{LSE}}}{\partial d_j} = r_j.$$

The sign is positive because increasing a distance makes the negative log marginal larger. If one instead writes logits or negative energies, the same identity appears with the opposite sign. All results below use the distance convention above.

For supervised classification, $h \in \mathbb{R}^C$ denotes class logits and

$$p = \text{softmax}(h)$$

denotes the output-layer class posterior. The label-clamped cross-entropy gradient is

$$\frac{\partial L_{\text{CE}}}{\partial h_j} = p_j - \nVdash\{j = y\}.$$

We reserve r for component responsibilities at an intermediate EM site and p for output class probabilities. We write $\nVdash\{\cdot\}$ for the indicator function and $\mathbf{1}$ for the all-ones vector; these are distinct objects despite the similar glyphs.

The supervised model has two maps of interest. The first map, W_1, sends the input to component energies. Its rows are treated as private component parameters in the clean single-site theory. The second map, W_2, sends component coordinates to class logits. It is the learned dense component-to-class map, or *corridor*, through which the supervised gradient is pulled back. The largest singular value of this corridor is written $\sigma_{\max}(W_2)$.

The NegLogSoftmin calibration used in the experiments is

$$\text{NLS}(d)_j = d_j + \log \sum_k \exp(-d_k).$$

It preserves pairwise distance differences and satisfies

$$\exp(-\,\text{NLS}(d)_j) = r_j.$$

Thus it converts raw distances into calibrated distances whose exponentials are responsibilities.

Volume control consists of an anti-collapse variance penalty L_{var} and an anti-redundancy decorrelation or total-correlation penalty L_{tc}. We write

$$L_{\text{aux}}(d) = L_{\text{LSE}}(d) + L_{\text{var}}(d) + L_{\text{tc}}(d)$$

for the local auxiliary EM objective when all terms are active. Unless otherwise stated, λ denotes the total auxiliary weight in the supervised objective,

$$L_{\text{total}} = L_{\text{CE}} + \lambda L_{\text{aux}}.$$

In the early ablation experiments, $\lambda_{\text{reg}} = 0.001$ unless noted.

Finally, we use the spectral norm of the LSE Hessian in distance coordinates,

$$\|H_d L_{\text{LSE}}\|_2,$$

as a site-level sharpness diagnostic. Since the LSE Hessian is bounded above by $1/2$, we refer to proximity to this bound as the $1/2$-*saturation diagnostic*. Saturation indicates sharp two-way component competition at the site; values far below $1/2$ indicate diffuse responsibilities.

Symbol	Meaning
x	input example
K	number of intermediate components
C	number of output classes
d_j	energy/distance of component j; lower is better
$r = \mathrm{softmax}(-d)$	responsibility vector over components
h	output logits
$p = \mathrm{softmax}(h)$	output class posterior
W_1	input-to-component map; private rows in the clean theory
W_2	component-to-class corridor; shared learned dense map
$\sigma_{\max}(W_2)$	largest singular value of the corridor
$\mathrm{NLS}(d)$	NegLogSoftmin calibrated distances
L_{LSE}	local log-sum-exp objective
L_{var}	variance penalty, anti-collapse term
L_{tc}	decorrelation/total-correlation penalty, anti-redundancy term
L_{aux}	local EM auxiliary objective
λ	weight on L_{aux} in supervised training
H_d	Hessian with respect to the distance coordinates d

Table 1: Notation used throughout the paper.

2 Introduction

Log-sum-exp objectives are everywhere in neural networks. Softmax classification, attention, contrastive losses, and energy-based objectives all contain the same exponentiation-and-normalization geometry. In distance or energy coordinates, that geometry has a simple consequence: gradients are responsibilities. If

$$L_{\mathrm{LSE}}(d) = -\log \sum_j \exp(-d_j), \qquad r = \mathrm{softmax}(-d),$$

then

$$\frac{\partial L_{\mathrm{LSE}}}{\partial d_j} = r_j.$$

The posterior responsibility of component j is not an auxiliary variable computed outside the optimizer. It is the gradient supplied to gradient descent.

This identity gives a precise sense in which standard objectives can implement implicit generalized EM. The E-step appears as a normalized gradient over components; the parameter update plays the role of a generalized M-step. Prior work established this identity and showed that it is prescriptive in a single-layer decoder-free encoder [12, 13]. There, an LSE objective with volume control learned structured prototype-like features, while removing parts of volume control produced the expected mixture failures: dead components, collapse, and redundancy. The same model also exhibited an unusual optimizer signature: SGD was insensitive to learning rate over a wide range, Adam offered little advantage, and convergence time was roughly fixed.

Those results invite a tempting inference: if responsibility gradients flow through backpropagation, perhaps whole neural networks inherit EM's conditioning. This paper argues that the inference is false. The gradient-responsibility identity is local. It holds at the coordinates seen by an LSE or softmax objective. It does not imply that the same responsibility structure survives arbitrary learned composition on the way to earlier parameters.

2.1 The Boundary Question

The question is therefore not whether neural objectives contain EM-like interfaces. They do. The question is where the EM property lives and how far it travels.

There are two different properties at stake. The first is local mixture geometry. A hidden EM site can suffer the same degeneracies as a mixture model: components can die, collapse, or duplicate one another. Volume control is meant to repair these at-site failures by keeping components alive and distinct.

The second property is optimizer conditioning. In an isolated LSE site with private affine component parameters, the responsibility gradient reaches each component directly. The Hessian in distance coordinates is uniformly bounded by $1/2$, and the corresponding parameter-space curvature is bounded by a data constant. This explains why the single-layer model behaved differently from a typical neural network optimizer problem.

The central issue is whether the second property survives when the same site is embedded inside a supervised network. In the model studied here, the first layer creates component distances, a NegLogSoftmin (NLS) transform creates an intermediate EM site, and a learned dense map W_2 sends those component coordinates to supervised class logits. The cross-entropy gradient reaching the first layer is therefore pulled back through W_2 and added to the local EM gradient. That is the minimal setting in which to ask whether EM conditioning survives ordinary backpropagation composition.

2.2 Main Claim

The answer is that implicit EM is local, graded, and structural.

It is *local*: responsibility structure exists at LSE and softmax sites, and it extends cleanly to parameters only when the path preserves component identity. It does not automatically penetrate learned dense maps.

It is *graded*: the relevant variable is the measured share of EM-structured gradient reaching the parameters. A fully connected supervised graph can behave like an isolated EM site when the local EM gradient dominates, and a visible EM site can behave like an ordinary supervised network when the cross-entropy path dominates.

It is *structural*: failures divide into at-site failures and corridor failures. At-site failures are repaired by local volume control. Corridor failures arise when a responsibility gradient is mixed through shared learned maps before reaching the parameters it is supposed to condition.

This is the neural version of a classical EM fact. EM is cleanest when the expected complete-data objective separates over component-private parameters. When components share parameters, the M-step becomes coupled and the clean per-component update is lost. Backpropagation through a learned dense corridor is parameter sharing in this sense. It creates a foreign contribution at the earlier parameters whose curvature is no longer controlled by the local LSE bound.

That structural claim must be separated from dominance. Sharing explains why the cross-entropy contribution is not EM-conditioned; dominance explains when that contribution controls the update. The gradient at W_1 is a sum,

$$\nabla_{W_1} L_{\mathrm{CE}} + \lambda \nabla_{W_1} L_{\mathrm{aux}}.$$

If the CE term dominates, the effective landscape is the composed supervised landscape. If the local auxiliary term dominates, the same graph can recover the single-site optimizer signature. This is why the claim is not that connectivity alone destroys EM conditioning. The claim is that learned composition creates a non-EM contribution, and the measured gradient ratio decides whether it wins.

2.3 Results in Brief

The theory gives the local boundary. For an LSE objective in distance coordinates, the gradient is a simplex-valued responsibility vector and the Hessian has spectral norm at most 1/2. With private affine component parameters, this yields a parameter-independent curvature bound. When the site is composed behind a learned dense map, the leading (Gauss–Newton) term of the cross-entropy Hessian is the conjugated form

$$J^\top W_2^\top (\mathrm{diag}(p) - pp^\top) W_2 J,$$

whose norm scales with the learned spectrum $\sigma_{\max}(W_2)^2$, so the clean parameter-independent guarantee is forfeited.

The experiments support the split in this controlled setting. Volume control transfers to an intermediate supervised site: it eliminates dead units, prevents collapse, and reduces redundancy across widths. But the optimizer conditioning does not transfer through ordinary supervised composition. In the full supervised model, SGD is again learning-rate sensitive and Adam is again useful.

The causal experiment identifies the mechanism. At the standard auxiliary weight, the CE path dominates the weighted local EM path by roughly 67–95× late in training, while the two gradients show no detectable alignment. The local LSE curvature remains bounded by 1/2, but the composed CE-path curvature is larger, non-monotone, and parameter-dependent. Cutting the CE path to W_1 with stop-gradient restores learning-rate insensitivity. Increasing the EM weight until the EM path dominates restores the same behavior without cutting the graph. Conditioning follows dominance, not mere connectivity.

2.4 Contributions

This paper makes the following contributions.

1. It proves a conditioning contrast between private-parameter LSE objectives and LSE sites composed behind learned dense maps.
2. It identifies which operations preserve the responsibility-gradient invariant and which destroy it.
3. It separates at-site failures from structural corridor failures.
4. It shows that volume control transfers to intermediate supervised EM sites and repairs dead or redundant components.
5. It shows that the single-site optimizer signatures of implicit EM do not survive ordinary dense supervised composition.
6. It causally restores the conditioning signature by stop-gradient and by EM dominance, showing that the mechanism is gradient dominance rather than graph connectivity alone.
7. It introduces the 1/2-saturation diagnostic as a local measure of mixture sharpness at an EM site.

3 Related Work and Positioning

This paper connects several literatures that are usually treated separately: EM as an optimization algorithm, classical EM failure modes, local learning in neural networks, auxiliary-loss interference, and discriminative versus generative training. The common thread is not that neural networks are mixture models. It is that log-sum-exp objectives create responsibility-shaped gradients, and the usefulness of those gradients depends on whether the responsibility structure reaches the parameters intact.

3.1 EM as an Optimization Object

The expectation-maximization algorithm was introduced as a method for maximum likelihood estimation with incomplete data [4]. Its classical form alternates between computing posterior responsibilities and maximizing an expected complete-data objective. The free-energy view of EM [10] makes this coordinate-ascent structure explicit and clarifies why EM-like updates can be studied as optimization steps rather than only as a probabilistic modeling procedure.

The closest optimization precedent for this paper is the analysis of EM as a preconditioned gradient method [15]. That line of work already treats EM's behavior as a consequence of the geometry of the responsibilities and the missing information in the model. Our contribution is different in two ways. First, we show that standard neural objectives instantiate the responsibility gradient algebraically at log-sum-exp sites. Second, we draw the boundary of the resulting conditioning guarantee: it survives private local parameterizations, but ordinary learned composition forfeits the parameter-independent bound.

The uniform curvature constant used in our theory is not itself new. It is the same multinomial-logistic Hessian geometry bounded by Bohning [2], and it can also be read through the elementary variance bound used in the proof of Lemma 1. The novelty is not the $1/2$ bound in isolation. The novelty is its address: in an implicit-EM neural objective, the bound lives at the local distance interface, and the question is whether backpropagation preserves or destroys it.

3.2 Classical EM Failure Modes

Classical EM is cleanest when the expected complete-data objective separates over component-private parameters. Responsibilities assign data to components, and each component receives its own M-step. This separation is what makes the algorithm interpretable and often well behaved.

The classical weaknesses of EM appear when this separation is lost or when the component geometry degenerates. Shared parameters produce coupled M-steps. Generalized or non-separable M-steps no longer give independent component updates. Mixture likelihoods can develop singular components, and hard or nearly hard assignments can leave components with little chance of recovery. Diffuse responsibilities, conversely, produce weak specialization and slow progress.

This paper uses these classical failure modes as a structural analogy: at-site volume-control failures are the local analogue of mixture degeneracy, and dense backpropagation failures are the analogue of the coupled, non-separable M-step that arises when components share parameters. Section 5 makes this structural parallel explicit and testable, without claiming that supervised cross-entropy is literally an EM complete-data objective.

3.3 Generative and Discriminative Training

The generative–discriminative distinction [11] is the high-level contrast a skeptic will invoke: local EM-like objectives are generative, cross-entropy is discriminative, so of course they differ. Our result is more specific. Within one architecture, holding data and the local site fixed and varying only how CE reaches W_1, we measure the mechanism—the bounded LSE curvature, the larger parameter-dependent CE-path curvature, the gradient dominance, and its causal reversal under stop-gradient. The contribution is not that two objectives differ but the identification of the corridor through which the discriminative objective destroys local EM conditioning.

3.4 Local and Decoupled Learning

The design implication of our theory is local: if an EM site is meant to keep its optimization structure, it needs an objective that reaches its parameters directly, or a mechanism that prevents a foreign gradient from dominating it. This places the work near layer-wise and decoupled learning methods.

Deep belief networks and greedy layer-wise pretraining demonstrated that useful representations can be learned by protected local objectives [8]. Later work on greedy layer-wise learning [1], decoupled neural interfaces and synthetic gradients [9], and Forward-Forward [7] also explores alternatives to fully end-to-end credit assignment.

Our contribution is not a new layer-wise training algorithm. It is a mechanism for why protected local objectives can preserve structure that ordinary backpropagation may not. In the implicit-EM setting, the protected object is a responsibility gradient at a local LSE site. If that gradient reaches private component parameters, the site inherits a parameter-independent curvature bound. If the same site is trained primarily by a dense foreign corridor, that bound no longer governs the update.

3.5 Auxiliary Losses, Gradient Dominance, and Interference

Training with both a main CE loss and a local auxiliary EM loss places the work near multi-task learning and gradient-interference methods: PCGrad [16] modifies conflicting task gradients, GradNorm [3] balances task weights, and gradient starvation [14] studies one signal dominating another. We do not propose a balancing method; the gradient ratio is instead our causal variable for whether EM conditioning is visible at W_1. Notably, the measured cosines show the EM signal is *drowned*, not opposed—distinguishing this from the conflicting-gradient regime that gradient surgery targets.

3.6 Loss Landscapes and Basin Structure

The paper also touches the loss-landscape literature through the proposed basin and interpolation analyses. Linear mode connectivity shows that networks trained in apparently different basins can often be connected by low-loss paths [6]. Permutation invariance is a central complication in such analyses: component or neuron relabeling can create artificial barriers if models are interpolated without alignment [5].

Those tools are useful here because the EM-dominated and CE-dominated regimes can produce representations with different task alignment and different local geometry. A basin analysis asks whether discriminative fine-tuning refines the EM solution or moves to a distinct region after hidden-unit alignment. This is supporting evidence rather than the main claim; the central causal evidence is the stop-gradient and dominance experiment.

3.7 The Framework's Own Lineage

This paper builds directly on the implicit-EM identity and the decoder-free volume-control model. The first established that log-sum-exp distance objectives produce responsibility gradients [12].

The second showed that LSE plus volume control learns structured prototype-like features in a single-layer unsupervised setting [13]. The present paper asks the next boundary question: which of those properties survive when the EM site is inserted into a supervised neural network?

The answer is split. Volume control transfers to the intermediate site and repairs dead or redundant components. The single-site optimizer conditioning does not transfer through ordinary dense composition unless the local EM path is isolated or made dominant. This distinction is the paper's contribution to the broader implicit-EM program.

3.8 Scope

This is not a general theory of deep-network optimization, Hessian spectra, or generalization. The claims are narrower. They concern log-sum-exp or softmax sites, the responsibility gradients those sites create, and the corridors through which those gradients reach parameters. Within that scope, the paper gives both a formal sufficient condition for preserving EM conditioning and an empirical account of how ordinary supervised composition destroys it.

4 Background: Where EM Lives

This section fixes the conceptual object studied in the rest of the paper. An implicit-EM site is not any hidden layer with a softmax nearby, and it is not an entire deep network. It is a local exponentiation-and-normalization interface at which an objective sees distances, energies, or logits and therefore produces a responsibility-shaped gradient. The theory section asks when this local structure survives the pullback to parameters.

4.1 Responsibilities as Gradients

The starting point (Section 2) is the algebraic identity $\partial L_{\mathrm{LSE}}/\partial d_j = r_j$: the gradient of an LSE distance objective is a posterior responsibility vector—nonnegative, summing to one, assigning each component a fractional share of the update. Gradient descent on such an objective performs an implicit E-step, with responsibilities appearing as the gradient rather than as auxiliary variables.

The identity is purely algebraic; it assumes no particular parameterization, architecture, or optimizer. Everything in this paper turns on what happens *next*: the responsibility vector may reach private component parameters directly, or it may be mixed with other gradients and pulled through learned maps before it updates earlier weights.

4.2 Cross-Entropy Is a Label-Clamped Responsibility Interface

Supervised cross-entropy has the same exponentiation-and-normalization interface, with the label clamping one component: its logit gradient is $p_c - \mathbb{1}\{c = y\}$ (Section 1), removing probability mass from the true class and redistributing it according to the model posterior.

This observation is useful but also dangerous. It is correct that the output loss supplies responsibility-shaped geometry at the output logits. It is *not* correct to conclude that every earlier hidden layer is itself doing EM. The gradient reaching an earlier layer has already passed through the head Jacobian and any intervening nonlinearities; it need not be nonnegative, sum to one, or remain aligned with hidden components.

4.3 Local EM Sites Inside a Network

The supervised architecture studied here deliberately creates an intermediate site. The first map produces component distances $d_1, \ldots, d_K$, and the NegLogSoftmin transform (Section 1) calibrates them so that $\exp(-\operatorname{NLS}(d)_j) = r_j$, giving the hidden layer an explicit distance-to-responsibility interface rather than relying on the output loss to impose structure indirectly.

The distinction between an output interface and an intermediate site is central. At the output, the supervised objective owns the softmax. At the hidden layer, the local site exists only because we add a local auxiliary objective. Without such an objective, the hidden coordinates may be useful features, but there is no reason for their updates to be posterior responsibilities over hidden components.

4.4 Volume Control

An LSE competition term by itself does not prevent the standard degeneracies of mixture models. Components can collapse, become redundant, or receive essentially no responsibility. The decoder-free encoder study addressed these failures with volume control: a variance term prevents dead or collapsed components, and a decorrelation term prevents all components from moving together. In the local auxiliary objective $L_{\text{aux}} = L_{\text{LSE}} + L_{\text{var}} + L_{\text{tc}}$ (Section 1), the variance term is the anti-collapse part and the decorrelation term the anti-redundancy part. Together they act as a volume barrier, encouraging components to occupy distinct directions rather than concentrate in the same small region—the neural analogue of the log-determinant in a Gaussian mixture.

The first empirical question is whether these local repairs still work inside a supervised network. They do: Section 7 shows that volume control eliminates dead units and reduces redundancy at the intermediate site. The second question is harder: whether the optimizer conditioning observed in the single-layer implicit-EM model transfers through the supervised head. Sections 8 and 9 show that it does not, except when the local EM path is isolated or made dominant.

4.5 Locality Versus Inheritance

The main conceptual error this paper rules out is inheritance. The gradient-responsibility identity guarantees a local structure at the coordinates seen by an LSE or softmax objective. It does not guarantee that the whole network inherits EM's conditioning. Backpropagation can transport the gradient, but transport is not preservation. Dense learned maps can mix signs and components; objective sums can add unrelated gradients; gates can turn small responsibilities into exactly zero updates.

Thus there are two different claims one might make about a hidden layer. A weak, correct claim is that loss geometry at downstream softmax sites shapes hidden representations through backpropagation. A strong, generally false claim is that the hidden layer itself receives a posterior responsibility update. The rest of the paper draws the boundary between these claims.

4.6 Two Failure Classes

The experiments are organized around two failure classes. At-site failures occur at a local EM interface: dead components, collapsed variance, redundant features, and diffuse or poorly calibrated responsibilities. These are the failures volume control is designed to repair.

Structural failures occur when a responsibility gradient must travel through a shared learned corridor before reaching the parameters it is supposed to condition. In the supervised model, this corridor is the component-to-class map W_2. The CE gradient reaching W_1 is pulled through W_2

and combined with the local auxiliary gradient. When this foreign path dominates, the update at W_1 no longer has the private, per-component structure of an EM M-step.

This split sets up the theory. The next section proves that private LSE sites inherit a parameter-independent curvature bound, while composition through a learned dense map forfeits that guarantee. The experiments then show the same boundary from both sides: local volume control transfers, but optimizer conditioning does not survive ordinary composition.

5 Theory: The Conditioning Boundary

The gradient-responsibility identity is local: it is a statement about the coordinates d at an exponentiation-and-normalization site. The question for a neural network is what happens after this responsibility vector is pulled back to parameters. This section isolates the condition under which the local curvature bound survives that pullback, and the condition under which it is forfeited.

5.1 The Simplex Invariant

Lemma 1 (Simplex invariant and curvature bound)**.** *For*

$$L_{\mathrm{LSE}}(d) = -\log \sum_{j=1}^{K} \exp(-d_j), \qquad r = \mathrm{softmax}(-d),$$

the gradient $\nabla_d L_{\mathrm{LSE}} = r$ lies on the probability simplex. The Hessian is

$$\nabla_d^2 L_{\mathrm{LSE}} = rr^\top - \mathrm{diag}(r).$$

Equivalently, the negative Hessian is the categorical covariance matrix

$$C(r) = \mathrm{diag}(r) - rr^\top \succeq 0, \qquad \|C(r)\|_2 = \|\nabla_d^2 L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}.$$

Proof. The gradient calculation gives

$$\frac{\partial L_{\mathrm{LSE}}}{\partial d_j} = \frac{\exp(-d_j)}{\sum_k \exp(-d_k)} = r_j.$$

Differentiating r with respect to d gives

$$\frac{\partial r_j}{\partial d_k} = -r_j(\mathbb{1}\{j=k\} - r_k),$$

so $\nabla_d^2 L_{\mathrm{LSE}} = rr^\top - \mathrm{diag}(r)$. For any vector v,

$$v^\top C(r) v = \sum_j r_j v_j^2 - \left(\sum_j r_j v_j\right)^2 = \mathrm{Var}_{j\sim r}(v_j).$$

By Popoviciu's inequality,

$$\mathrm{Var}_{j\sim r}(v_j) \le \frac{1}{4}(\max_j v_j - \min_j v_j)^2.$$

For $\|v\|_2 = 1$, the squared range is at most 2, hence $v^\top C(r) v \le 1/2$. Taking the supremum over unit vectors proves the bound. □

The important feature is not merely boundedness, but *where the bound lives.* In distance coordinates it is absolute: no parameter value, scale, or training time appears. The gradient is a normalized conditional expectation, and the curvature is the covariance of that conditional distribution.

5.2 Private Parameters Inherit the Bound

Proposition 1 (Conditioning under private affine energies)**.** *Let each component energy be a private affine map*

$$d_j = w_j^\top x + b_j,$$

and let $\theta = (w_1, b_1, \ldots, w_K, b_K)$. For a single example,

$$\|\nabla_\theta L_{\mathrm{LSE}}\|_2 \le \sqrt{\|x\|^2 + 1},$$

and

$$\|\nabla_\theta^2 L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}(\|x\|^2 + 1).$$

If biases are omitted from the norm or absorbed into an augmented input, this is the bound $\|\nabla_\theta^2 L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}\|\tilde{x}\|^2$. Under batch-mean training,

$$\|\nabla_\theta^2 L_{\mathrm{LSE}}\|_2 \le \frac{1}{2}\mathbb{E}\|\tilde{x}\|^2,$$

a constant determined by the data and independent of the current parameters.

Proof. Let $\tilde{x} = (x, 1)$ absorb the bias and write $d_j = \theta_j^\top \tilde{x}$. Since d is linear in the private parameter blocks, there is no second-order term from the map $\theta \mapsto d$. The gradient block for component j is $r_j \tilde{x}$, so

$$\|\nabla_\theta L_{\mathrm{LSE}}\|_2^2 = \sum_j r_j^2 \|\tilde{x}\|^2 \le \|\tilde{x}\|^2.$$

The Hessian has blocks

$$\left(\nabla_\theta^2 L_{\mathrm{LSE}}\right)_{jk} = \left(rr^\top - \mathrm{diag}(r)\right)_{jk} \tilde{x}\tilde{x}^\top,$$

or, up to sign, $C(r) \otimes \tilde{x}\tilde{x}^\top$. The spectral norm of a Kronecker product is the product of spectral norms, and Lemma 1 gives the factor $1/2$. □

Proposition 1 is the formal version of the single-site conditioning claim. The smoothness constant is fixed before training starts. Whatever step-size range is allowed by the data-dependent constant does not shrink as the parameters move, and there is no learned anisotropy for an adaptive optimizer to correct. This is the mechanism behind the optimizer signature observed in the single-layer implicit-EM model.

This clean bound applies to the LSE contribution. The full auxiliary objective used experimentally also contains variance and decorrelation terms, whose curvature is not covered by Proposition 1. We therefore use the proposition as the local conditioning baseline: it identifies the parameter-independent part supplied by the responsibility site, while the experiments measure the behavior of the full auxiliary objective in the composed model.

Proposition 2 (Smooth monotone kernels)**.** *Let*

$$d_j = \phi(w_j^\top x + b_j)$$

with private component parameters, $\phi' \geq 0$, $|\phi'| \leq B_1$, and $|\phi''| \leq B_2$. Then the parameter-space curvature of $L_{\mathrm{LSE}}(d(\theta))$ is bounded by a constant multiple of $\|\tilde{x}\|^2$ that depends only on B_1 and B_2, not on the current parameters.

Proof. The chain rule gives two terms:

$$\nabla_\theta^2 L = J^\top \nabla_d^2 L J + \sum_j \frac{\partial L}{\partial d_j} \nabla_\theta^2 d_j.$$

The first term is bounded by Lemma 1 and $\|J\|^2 \leq B_1^2 \|\tilde{x}\|^2$. The second term is block diagonal across components; since $\partial L/\partial d_j = r_j$ and $\sum_j r_j = 1$, its norm is bounded by $B_2\|\tilde{x}\|^2$. Thus the total curvature is bounded by $(B_1^2/2 + B_2)\|\tilde{x}\|^2$ up to the harmless convention for including biases. □

Softplus is the main smooth case in this paper. ReLU should be read separately: its curvature bounds hold almost everywhere, but it introduces a zeroth-order absorbing-state pathology. When a unit is gated off, its gradient can become exactly zero, and the unit may not recover. That failure is not a violation of the curvature bound; it is the neural version of hard-assignment cluster death.

5.3 Composition Forfeits the Uniform Bound

We now compose the same local site behind a learned dense map. Let

$$y = \mathrm{NLS}(d), \qquad h = W_2 y, \qquad L_{\mathrm{CE}} = -h_c + \log \sum_\ell \exp(h_\ell),$$

and $p = \mathrm{softmax}(h)$. The Jacobian of $y = \mathrm{NLS}(d)$ is

$$J = \frac{\partial y}{\partial d} = I - \mathbf{1} r^\top,$$

because $\partial \log \sum_k \exp(-d_k)/\partial d_j = -r_j$.

Proposition 3 (Composition forfeits the uniform bound)**.** *For the composed supervised path above, the Hessian of L_{CE} with respect to the intermediate distances d decomposes as*

$$\nabla_d^2 L_{\mathrm{CE}} = \underbrace{J^\top W_2^\top \left(\mathrm{diag}(p) - pp^\top\right) W_2 J}_{\textit{Gauss–Newton term}} + \underbrace{\sum_l \frac{\partial L_{\mathrm{CE}}}{\partial y_l} \nabla_d^2 y_l}_{\textit{NLS second-order term}}.$$

The Gauss–Newton term is bounded by $\frac{K}{2}\,\sigma_{\max}(W_2)^2$, a quantity that depends on the learned *spectrum of W_2; the uniform, parameter-independent bound of Lemma 1 therefore no longer controls the composed curvature. We bound the Gauss–Newton term explicitly and do not claim a parameter-independent bound on the NLS second-order term; the point is only that composition introduces a $\sigma_{\max}(W_2)^2$ dependence absent from the isolated site, which the empirical curvature of Section 9 then shows to be large in practice.*

Proof. The label clamp $-h_c$ in L_{CE} is linear in h, so it contributes to the gradient $p - e_c$ but not to the second derivative. The CE Hessian with respect to logits is therefore the softmax Hessian

$$H_h L_{\mathrm{CE}} = \mathrm{diag}(p) - pp^\top,$$

independent of the label. Pulling this term back through the linear map $h = W_2 y$ and the NLS Jacobian $J = \partial y/\partial d$ gives the displayed Gauss–Newton term $J^\top W_2^\top(\operatorname{diag}(p) - pp^\top)W_2 J$; the remaining terms come from the second derivatives of $y = \mathrm{NLS}(d)$, weighted by the incoming gradient from the head.

To bound the leading term we need the two conjugating norms. First, the softmax Hessian satisfies $\|\operatorname{diag}(p) - pp^\top\|_2 \le 1/2$ exactly as in Lemma 1, so

$$\|W_2^\top(\operatorname{diag}(p) - pp^\top)W_2\|_2 \le \frac{1}{2}\sigma_{\max}(W_2)^2.$$

Second, we bound the NLS Jacobian $J = I - \mathbf{1}r^\top$, which is not a contraction. We compute $\|J\|_2$ in closed form. Writing

$$J^\top J = I - \mathbf{1}r^\top - r\mathbf{1}^\top + K\,rr^\top = I + M, \qquad M := -\mathbf{1}r^\top - r\mathbf{1}^\top + K\,rr^\top,$$

the perturbation M is supported on $\operatorname{span}\{\mathbf{1}, r\}$: for any v orthogonal to both $\mathbf{1}$ and r we have $Mv = 0$, so $J^\top J$ acts as the identity on that complement (eigenvalue 1). If $r = \mathbf{1}/K$, then J is the orthogonal projector onto $\mathbf{1}^\perp$, so the eigenvalues of $J^\top J$ are 1 with multiplicity $K-1$ and 0 once. Otherwise $\mathbf{1}$ and r are linearly independent. On $\operatorname{span}\{\mathbf{1}, r\}$, using $\mathbf{1}^\top\mathbf{1} = K$, $\mathbf{1}^\top r = 1$, and $r^\top r = \|r\|_2^2 =: s$, the operator M has matrix representation $\left(\begin{smallmatrix} -1 & -s \\ 0 & Ks-1 \end{smallmatrix}\right)$ in the basis $\{\mathbf{1}, r\}$, with eigenvalues -1 and $Ks - 1$. Hence the eigenvalues of $J^\top J$ are $\{1$ (multiplicity $K-2$), 0, $Ks\}$ away from the uniform point, with the same top eigenvalue formula at the uniform point. Thus

$$\|J\|_2^2 = \lambda_{\max}(J^\top J) = \max(1, Ks) = K\|r\|_2^2,$$

since $Ks \ge 1$ for a probability vector (as $\|r\|_2^2 \ge 1/K$). Because $\|r\|_2^2 \le \|r\|_1 = 1$ on the simplex, with equality only at a vertex,

$$\|J\|_2 = \sqrt{K}\,\|r\|_2 \le \sqrt{K},$$

attained exactly when r is one-hot and equal to 1 when $r = \mathbf{1}/K$. We keep this factor explicit rather than absorb it into an unnamed constant. Combining,

$$\left\|J^\top W_2^\top(\operatorname{diag}(p) - pp^\top)W_2 J\right\|_2 \le \|J\|_2^2 \cdot \tfrac{1}{2}\sigma_{\max}(W_2)^2 \le \tfrac{K}{2}\,\sigma_{\max}(W_2)^2.$$

The leading factor is $\sigma_{\max}(W_2)^2$, modulated by an NLS-Jacobian factor bounded by $K/2$ that depends only on the number of components. Two features distinguish this from Lemma 1: the bound scales with the *learned* spectrum of the corridor W_2, and it is therefore parameter-dependent and time-varying rather than a fixed data constant. The NLS factor is the (parameter-independent) price of calibration; the corridor factor is the source of the forfeited guarantee. This bounds the Gauss–Newton term only; we do not bound the NLS second-order term $\sum_l (\partial L_{\mathrm{CE}}/\partial y_l)\nabla_d^2 y_l$, which need not be small (it carries the incoming head gradient) and which the empirical measurements in Section 9 include. The claim is not that the full composed curvature is controlled by $\sigma_{\max}(W_2)^2$, but that composition introduces a $\sigma_{\max}(W_2)^2$ dependence that the isolated site does not have. □

Proposition 3 is deliberately a guarantee-forfeiture statement. It shows that the parameter-independent 1/2 bound of the local LSE site no longer controls the composed supervised path. It does *not* prove that the curvature must be large for every possible W_2 or every point in training. The empirical curvature measurements in Section 9 carry that claim for the studied networks: the composed CE-path curvature is larger, non-monotone, and parameter-dependent.

We state the clean proposition without LayerNorm. The measurements are taken on the actual model, including LayerNorm, so its effect is included in the reported curvature. A proof with LayerNorm would insert another per-sample Jacobian and additional input-dependent terms; it would not restore the parameter-independent LSE guarantee.

Scope of the claim. The propositions above establish sufficient conditions for preserving the local EM conditioning guarantee: private component parameters, an unmixed LSE objective, and monotone per-coordinate kernels. They also show that an intervening shared, sign-indefinite learned map forfeits the uniform guarantee. They do not prove necessity for arbitrary architectures and do not provide a convergence theorem for the composed system.

5.4 Operations That Preserve or Destroy the Invariant

The preceding propositions can be summarized by how operations treat the simplex-valued responsibility gradient.

Preserving operations. Private per-component parameters preserve the alignment between responsibility r_j and parameter block θ_j. Per-coordinate monotone maps preserve sign and component identity, changing only local scales. Smooth bounded kernels preserve parameter-independent curvature up to constants. Stochastic maps are a partial first-order preservation class: if A is row-stochastic, then pulling a nonnegative unit-mass vector back through $A^\top$ preserves nonnegativity and total mass because $\mathbf{1}^\top A^\top g = (A\mathbf{1})^\top g = \mathbf{1}^\top g$.

Destroying operations. Dense sign-indefinite learned maps do not preserve nonnegativity, normalization, or component alignment. Objective mixing replaces a single posterior gradient with a sum of gradients from different objectives; the sum is generally not the posterior of any one latent-variable model. Hard gates such as ReLU can turn small responsibility into exactly zero gradient, creating absorbing states. Shared lower parameters couple the component updates and destroy the separated M-step structure.

5.5 Classical EM Connection

The last point is the link to classical EM. In the clean case, the E-step produces responsibilities and the expected complete-data objective separates:

$$Q(\theta) = \sum_{j=1}^{K} Q_j(\theta_j).$$

Each component owns its own parameters, and the M-step decouples into independent per-component optimizations. This separation is the source of EM's algorithmic structure.

If all components depend on a shared lower map, the separation disappears. Writing the lower parameters as ψ, the complete-data terms have the form $Q_j(\theta_j, \psi)$, so

$$Q(\theta, \psi) = \sum_j Q_j(\theta_j, \psi)$$

is no longer separable in ψ. The M-step becomes a single coupled optimization problem. Backpropagation through shared lower layers is the neural version of this classical shared-parameter failure, and the conjugated Hessian in Proposition 3 is its coordinate-space fingerprint.

This structural story must be reconciled with the graded empirical result. The claim is not that sharing alone always destroys the effective conditioning of the full update. Rather, sharing creates a foreign CE-path contribution whose curvature is no longer controlled by the local LSE bound. The update at W_1 contains both this contribution and the local EM contribution:

$$\nabla_{W_1} L_{\text{total}} = \nabla_{W_1} L_{\text{CE}} + \lambda \nabla_{W_1} L_{\text{aux}}.$$

Classical EM failure	Why EM weakens	Neural incarnation
Shared component parameters	Q stops separating; the M-step is coupled	Backpropagation through shared lower maps
Non-separable / generalized EM	No closed-form independent M-step	Component energies computed by deep maps
Singular components	Variance collapses and likelihood becomes unbounded	LSE-only collapse without volume control
Hard assignment	Zero-responsibility components cannot recover	Dead ReLU units
High missing information	Diffuse responsibilities slow convergence	Low 1/2-saturation and weak competition

Table 2: Classical EM failure modes and their neural counterparts.

When the CE contribution dominates, the effective landscape is the non-separable shared-parameter landscape. When the EM contribution dominates, the same graph can behave like the isolated EM site. Thus the structural result explains why the foreign term is ill-conditioned, while the dominance result explains when that term matters.

6 Experimental Setup

The experiments use deliberately small supervised networks. The goal is not to improve MNIST accuracy, but to make the location and failure mode of implicit EM measurable. The architecture is kept fixed across experiments except for the controlled ablations described below.

6.1 Model Family

The base model is a two-layer MNIST classifier with an intermediate component site:

$$x \in \mathbb{R}^{784} \to \mathrm{Linear}(784, K) \to \mathrm{ReLU} \to d \to \mathcal{A}(d) \to \mathrm{Linear}(K, 10) \to \mathrm{LayerNorm} \to L_{\mathrm{CE}}.$$

Here K is the number of intermediate components, d is the vector of intermediate energies, and $\mathcal{A}$ is the ablated intermediate transform. Depending on the experiment, $\mathcal{A}$ is either the identity map or NegLogSoftmin:

$$\mathrm{NLS}(d)_j = d_j + \log \sum_k \exp(-d_k).$$

When NLS is present, the site has the LSE partition function and the competitive Jacobian needed for a local implicit-EM interpretation. When NLS is absent, the same downstream classifier receives the raw ReLU-produced coordinates.

The second linear map, W_2, is the learned dense corridor from intermediate components to class logits. It is the map whose pullback destroys the uniform conditioning guarantee in Proposition 3. LayerNorm is kept in the empirical model because it is part of the trained supervised baseline; the clean theory omits it and the curvature measurements include its effect.

6.2 Local Auxiliary Objective

The local EM objective is applied to the intermediate distances d. In the causal experiment it is

$$L_{\mathrm{aux}}(d) = L_{\mathrm{LSE}}(d) + L_{\mathrm{var}}(d) + L_{\mathrm{tc}}(d).$$

The variance term L_{var} is the anti-collapse term: it penalizes components whose activations have too little variance over the batch. The decorrelation term L_{tc} is the anti-redundancy term: it penalizes correlated component responses. Together they implement the volume-control mechanism tested in the decoder-free encoder study.

The supervised objective is

$$L_{\text{total}} = L_{\text{CE}} + \lambda L_{\text{aux}},$$

with $\lambda_{\text{reg}} = 0.001$ in the initial ablation and capacity experiments unless otherwise stated. In Experiments 1–3, the auxiliary loss uses the volume-control terms used in those reports. In Experiment 4, the LSE term is included explicitly so that the stop-gradient arm trains W_1 by the same local EM objective as the single-layer model.

6.3 Experiment 1: Volume-Control Ablation

Experiment 1 tests which pieces of the intermediate EM site are needed. The architecture uses $K = 25$ components, MNIST, 50 training epochs, Adam with learning rate 10^{-3}, $\lambda_{\text{reg}} = 0.001$, and 10 seeds $(42, \ldots, 51)$. Metrics are averaged over the final 10 epochs.

The five configurations are:

Configuration	NLS	L_{var}	L_{tc}
Baseline	–	–	–
NLS only	✓	–	–
NLS + Var	✓	✓	–
NLS + Var + Decorr	✓	✓	✓
Var + Decorr only	–	✓	✓

This ablation separates three questions: whether supervision alone prevents intermediate collapse, whether competition without volume control is enough, and whether volume control is merely regularization or specifically stabilizes an EM site.

6.4 Experiment 2: Capacity Sweep

Experiment 2 tests whether the at-site repair persists across width. It compares the baseline against NLS plus full volume control at

$$K \in \{16, 25, 36, 49, 64\}.$$

Each model is trained on MNIST for 50 epochs with Adam at learning rate 10^{-3}, $\lambda_{\text{reg}} = 0.001$, and 5 seeds $(42, \ldots, 46)$. Metrics are again averaged over the final 10 epochs. This experiment is used to distinguish structural health from benchmark accuracy: volume control can eliminate dead units and reduce redundancy even when the fixed regularization weight is not accuracy-optimal at larger widths.

6.5 Experiment 3: Optimizer Sweep

Experiment 3 asks whether the single-site optimization signature survives ordinary supervised composition. It fixes the NLS + full-volume-control model with $K = 25$ and $\lambda_{\text{reg}} = 0.001$, and sweeps optimizer and learning rate:

$$\text{optimizer} \in \{\text{SGD}, \text{Adam}\}, \qquad \eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}.$$

Each setting is trained for 50 epochs on MNIST with 3 seeds $(42, 43, 44)$. The comparison is deliberately aligned with the single-layer implicit-EM signature: learning-rate sensitivity, Adam advantage, and whether lower auxiliary loss corresponds to better features.

6.6 Experiment 4: Mechanism and Causal Test

Experiment 4 measures the mechanism behind the optimizer result. The model is

$$x \to \text{Linear}(784, 25) \to \text{ReLU} \to d \to \text{NLS}(d) \to \text{Linear}(25, 10) \to \text{LayerNorm} \to L_{\text{CE}},$$

with

$$L_{\text{aux}}(d) = L_{\text{LSE}}(d) + L_{\text{var}}(d) + L_{\text{tc}}(d).$$

Four arms differ only in how the supervised and local objectives reach W_1:

Arm	Total loss	CE reaches W_1?
$\lambda = 0.001$ joint	$L_{\text{CE}} + 0.001L_{\text{aux}}$	✓
$\lambda = 0.03$ joint	$L_{\text{CE}} + 0.03L_{\text{aux}}$	✓
$\lambda = 1$ joint	$L_{\text{CE}} + L_{\text{aux}}$	✓
stop-gradient	$L_{\text{CE}}(\text{head}(y.\text{detach}())) + L_{\text{aux}}$	–

The stop-gradient arm keeps the same data, labels, architecture, and supervised head training, but prevents the CE gradient from reaching W_1. This makes it the causal control for the corridor.

For the conditioning sweep, each arm is trained with SGD at

$$\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$$

for 40 epochs and 3 seeds. Feature quality is measured after training by a fixed-protocol linear probe trained on the frozen intermediate distances d. This decouples the measured representation from how well the supervised head trained at a particular learning rate.

For gradient competition, the training loop measures

$$\|\nabla_d L_{\text{CE}}\|, \qquad \lambda\|\nabla_d L_{\text{aux}}\|, \qquad \cos(\nabla_d L_{\text{CE}}, \nabla_d L_{\text{aux}}).$$

These measurements are taken during Adam training at learning rate 10^{-3}, while the conditioning sweep uses SGD. We therefore report these ratios as a site-level gradient diagnostic rather than as a matched-optimizer measurement; the matched-SGD version is a strengthening check.

For curvature tracking, power iteration with exact Hessian-vector products estimates the spectral norm of the Hessian with respect to d for both the LSE term and the CE path through NLS and the head. Measurements use a fixed 512-example batch every 2 epochs, and $\sigma_{\max}(W_2)$ is recorded alongside the curvature estimates.

6.7 Metrics

Representation health is measured by four quantities. A dead unit is a component whose activation variance is effectively zero over the evaluation window. Minimum variance records the weakest component variance. Redundancy measures correlation or duplication among component responses. Responsibility entropy is computed from $r = \text{softmax}(-d)$ and measures how sharp the local component assignments are.

Optimization conditioning is measured by learning-rate spread, convergence-time stability, Adam versus SGD gaps, gradient dominance ratios, cosine similarity between CE and auxiliary gradients, and Hessian spectral norms with respect to the intermediate distances. For cosine measurements, the analysis should compare the observed values to the random-vector scale $1/\sqrt{K}$, not merely to zero. For convergence time, the reported LSE-stabilization metric is interpreted as an EM-site stability measure; in CE-dominated arms the classifier may train well even if the local LSE site does not stabilize.

6.8 Experimental Logic

The experiments follow the paper's taxonomy. Experiments 1–2 test at-site failures: collapse, dead units, and redundancy. Their question is whether local volume control repairs these failures inside a supervised network. Experiment 3 tests whether the single-site conditioning signature survives ordinary supervised composition. Experiment 4 separates structure from dominance: the dense corridor creates an ill-conditioned CE-path contribution, while the measured CE/EM gradient ratio determines whether that contribution controls the effective landscape.

7 Results I: At-Site Failures Are Repaired Locally (Experiments 1–2)

The theory separates two kinds of failure. Structural failures arise when a responsibility gradient is pulled through a shared learned corridor. At-site failures occur at the EM interface itself: components die, collapse, or become redundant. The first two experiments test the at-site half. They ask whether supervision supplies enough intermediate volume control, and whether the variance and decorrelation terms repair the same failures inside a supervised network that they repair in the single-layer model.

7.1 Supervision Does Not Supply Intermediate Volume Control

Experiment 1 ablates the intermediate EM site at $K = 25$. All models share the same supervised classifier after the intermediate layer; only NLS and the local volume-control terms are changed. Table 3 reports the summary over the final 10 epochs and 10 seeds.

Config	Dead Units	Min Var	Redundancy	Resp. Entropy	Accuracy
Baseline	1.40 ± 0.49	0.000 ± 0.000	20.5 ± 2.8	3.08 ± 0.02	96.09 ± 0.07
NLS only	1.20 ± 0.98	0.264 ± 0.408	24.9 ± 3.0	2.93 ± 0.02	96.14 ± 0.17
NLS + Var	0.00 ± 0.00	11.74 ± 3.08	187.0 ± 38.7	2.09 ± 0.09	95.75 ± 0.21
NLS + Var + Decorr	0.00 ± 0.00	7.84 ± 1.17	13.9 ± 0.5	2.46 ± 0.04	96.34 ± 0.11
Var + Decorr only	0.10 ± 0.30	3.55 ± 1.49	13.3 ± 0.9	2.65 ± 0.07	96.40 ± 0.10

Table 3: Volume-control ablation at $K = 25$. Redundancy is $\|\operatorname{Corr}(d) - I\|_F^2$ over the batch (larger is worse); dead units count components with near-zero activation variance. Accuracy is in percent. All values are mean $\pm$ standard deviation over seeds.

The baseline has minimum variance zero across all seeds and an average of 1.4 dead units out of 25. Supervised cross-entropy trains the output layer, but the gradient reaching the hidden layer through W_2 does not prevent intermediate collapse. This is the first at-site result: labels provide structure at the output interface, but not volume control for an intermediate component site.

7.2 Competition Without Volume Control Is Not Enough

NLS alone introduces the LSE partition function and the competitive Jacobian at the intermediate site. If competition by itself stabilized the representation, NLS-only should improve the health metrics. It does not. Accuracy is unchanged relative to the baseline, dead units remain, and redundancy increases from 20.5 to 24.9. The local EM geometry is present, but without volume control it has no reason to keep every component alive and distinct.

This matters for the paper's central distinction. The failure here is not that the EM identity is absent. The NLS site supplies the identity. The failure is that an EM site without volume control has the same degeneracies as an unregularized mixture model.

7.3 Partial Volume Control Is Worse Than None

The variance-only arm isolates the anti-collapse half of volume control. It works on the metric it directly targets: dead units fall to zero and minimum variance rises to 11.74. But it is the worst configuration overall. Redundancy explodes to 187.0, roughly nine times the baseline, and accuracy falls to 95.75%.

The mechanism is straightforward. A variance penalty forces every component to move, but without decorrelation the cheapest way for all components to have high variance is for them to move together. The representation becomes alive but redundant. The low responsibility entropy in this arm, 2.09, should not be read as healthy specialization: it is sharp competition among nearly duplicated components.

This is the cleanest replication of the decoder-free encoder ablation inside a supervised network. The diagonal and off-diagonal parts of volume control are not interchangeable. Anti-collapse without anti-redundancy can be worse than neither.

7.4 Full Volume Control Repairs the Site

Adding decorrelation fixes the variance-only pathology. The full NLS + Var + Decorr model has zero dead units, high minimum variance, and the lowest redundancy among the EM-site configurations. Redundancy falls to 13.9, and accuracy rises to 96.34% with a tight seed standard deviation.

The accuracy improvement is not the main claim. The main claim is that local volume control repairs local structural health. The full model keeps every component alive and distinct while preserving the calibrated responsibility structure introduced by NLS.

7.5 Volume Control Without NLS Is Good Regularization, Not the Same Mechanism

The Var + Decorr only arm is an important control. It matches the full EM-site model on accuracy and redundancy, showing that volume control is also a useful regularizer. But the representation differs: responsibility entropy is higher (2.65 versus 2.46), and minimum variance is lower (3.55 versus 7.84). Without NLS, the penalties improve feature geometry but do not create the same calibrated distance-to-responsibility interface.

This control keeps the interpretation honest. The paper is not claiming that every gain comes from a mysterious EM effect. Some gains are ordinary regularization. The EM-specific claim is narrower: when a local EM site exists, the same volume-control terms that stabilize mixture components stabilize the site's responsibilities and component geometry.

7.6 The Capacity Sweep: Structural Health Persists Across Width

Experiment 2 tests whether the ablation result is specific to $K = 25$. It compares the baseline against NLS plus full volume control for $K \in \{16, 25, 36, 49, 64\}$, using 5 seeds per width. Table 4 summarizes the key structural and accuracy metrics.

K	Config	Dead	Dead Frac.	Min Var	Redundancy	Accuracy
16	Baseline	1.0 ± 0.9	6.3%	0.097 ± 0.119	11.0 ± 2.2	95.19 ± 0.19
16	Full VC	0.0 ± 0.0	0.0%	5.22 ± 1.09	9.1 ± 0.9	95.32 ± 0.21
25	Baseline	1.4 ± 0.5	5.6%	0.000 ± 0.000	19.0 ± 1.3	96.07 ± 0.07
25	Full VC	0.0 ± 0.0	0.0%	7.76 ± 1.21	13.8 ± 0.5	96.35 ± 0.12
36	Baseline	2.4 ± 1.9	6.7%	0.030 ± 0.059	31.7 ± 5.1	96.63 ± 0.14
36	Full VC	0.0 ± 0.0	0.0%	9.51 ± 0.10	21.1 ± 1.5	96.60 ± 0.10
49	Baseline	2.5 ± 0.9	5.0%	0.000 ± 0.000	48.1 ± 3.6	97.04 ± 0.09
49	Full VC	0.0 ± 0.0	0.0%	12.65 ± 1.55	29.7 ± 0.5	96.87 ± 0.16
64	Baseline	4.1 ± 1.3	6.3%	0.000 ± 0.000	69.0 ± 7.2	97.29 ± 0.06
64	Full VC	0.0 ± 0.0	0.0%	15.33 ± 0.92	41.2 ± 0.8	97.00 ± 0.09

Table 4: Capacity sweep. Accuracy is reported in percent. Full VC denotes NLS plus variance and decorrelation penalties.

The dead-unit result is width-invariant. Every baseline width has dead units, and the dead fraction is stable at roughly 5–7%. Full volume control eliminates dead units at every width. This is not a small-architecture accident: the baseline routes around dead capacity, while volume control keeps all components active.

Redundancy also scales in the predicted direction. In the baseline, redundancy grows roughly linearly with width, from 11.0 at $K = 16$ to 69.0 at $K = 64$. Full volume control lowers redundancy at every width and sharply reduces the seed-to-seed variance. At $K = 64$, the redundancy standard deviation falls from 7.2 to 0.8.

7.7 Accuracy Is Secondary and Capacity-Dependent

The accuracy pattern is more nuanced. With the fixed λ_{reg} used in the sweep, volume control improves accuracy at $K = 16$ and $K = 25$, is neutral around $K = 36$, and slightly hurts at $K = 49$ and $K = 64$. This is not a failure of the structural claim. The fixed regularization weight is not calibrated per width, and the larger models can use correlated extra units as boundary refiners. A decorrelation penalty treats this cooperation as redundancy, even when it helps the task.

Thus the result should be read as two regimes. In the capacity-constrained regime, dead and redundant units are expensive; volume control improves both structure and accuracy. In the capacity-abundant regime, overparameterization can absorb dead units and exploit correlated features for supervised boundary refinement. Volume control still produces healthier component geometry, but the task accuracy trade-off depends on λ and width.

7.8 Conclusion of Results I

Experiments 1–2 establish the at-site half of the paper. The supervised gradient does not supply volume control to intermediate components. Competition without volume control is insufficient. Partial volume control can be pathological. Full volume control repairs collapse and redundancy locally, and the repair persists across width.

This is exactly the part of implicit EM that transfers through supervised networks: local mixture degeneracies can be fixed by local volume control. The next results ask whether the optimizer conditioning transfers as well.

8 Results II: Conditioning Does Not Survive Composition (Experiment 3)

Results I showed that volume control transfers to an intermediate supervised site: it eliminates dead units and reduces redundancy even though the network is trained for classification. Experiment 3 asks whether the optimization conditioning from the single-layer implicit-EM model transfers as well. The answer is no. Once the local site is embedded behind an ordinary supervised head, SGD becomes learning-rate sensitive again and Adam becomes useful again.

This experiment uses the full intermediate objective, NLS plus variance and decorrelation, with $K = 25$ and $\lambda_{\text{reg}} = 0.001$. We train with SGD and Adam over learning rates $\{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$ for 50 epochs on MNIST. Table 5 reports metrics averaged over epochs 41–50 and 3 seeds.

Opt.	LR	Acc.	CE	Reg.	Min Var	Redundancy
SGD	10^{-4}	83.83 ± 4.96	0.868	107.5	0.015	27.5
SGD	10^{-3}	93.69 ± 0.26	0.275	69.9	0.065	19.6
SGD	10^{-2}	95.63 ± 0.12	0.155	19.2	0.436	17.7
SGD	10^{-1}	96.23 ± 0.09	0.133	-34.3	4.01	16.2
Adam	10^{-4}	95.34 ± 0.33	0.170	24.6	0.324	18.0
Adam	10^{-3}	96.38 ± 0.11	0.133	-57.3	7.80	13.8
Adam	10^{-2}	96.37 ± 0.12	0.160	-158.3	423.2	11.9
Adam	10^{-1}	96.58 ± 0.05	0.179	-265.9	30628.4	11.9

Table 5: Optimizer sweep for the full intermediate objective. Accuracy is in percent. Metrics are averaged over the final 10 epochs and 3 seeds.

8.1 The Single-Site Optimizer Signature Disappears

The single-layer implicit-EM model had an unusual optimization signature: SGD was insensitive to learning rate over a wide range, Adam offered little advantage, and convergence time was largely fixed. Those properties do not survive ordinary supervised composition.

With SGD, accuracy spans 83.83% to 96.23% across the four learning rates, a 12.4 point range. At 10^{-4}, the model has not converged by epoch 50; at 10^{-3}, it reaches 93.69% but remains well below the best settings. The learning-rate axis is therefore active again. The local EM site and its volume control are present, but they do not make the composed supervised landscape learning-rate insensitive.

This is the empirical counterpart of Proposition 3. Composition does not prove that curvature must be large at every point, but it removes the parameter-independent guarantee that explained the single-site behavior. Experiment 3 shows that, in this supervised network, the missing guarantee matters.

8.2 Adam Becomes Useful Again

Adam clearly improves the low-learning-rate regime. At learning rate 10^{-4}, Adam reaches 95.34%, close to SGD at 10^{-2} and far above SGD at the matched learning rate. At 10^{-3}, Adam reaches 96.38%, matching or slightly exceeding SGD at 10^{-1}. The CE loss tells the same story: Adam finds low supervised loss efficiently at learning rates where SGD is still undertrained.

This is not a contradiction of Results I. Volume control is still repairing the intermediate site. The point is narrower: the optimizer no longer sees the private, responsibility-weighted landscape of the single-layer EM model. The gradient reaching W_1 is the sum of a local auxiliary contribution and a supervised contribution pulled back through the learned head. That composed gradient behaves like an ordinary deep-network gradient, and adaptive scaling is useful.

8.3 The Local Site Still Responds to Volume Control

Although the conditioning signature disappears, the local volume-control objective is not inert. As learning rate increases, the structural metrics move in the expected direction. For SGD, minimum variance rises from 0.015 to 4.01 and redundancy falls from 27.5 to 16.2. For Adam, redundancy falls to 13.8 at 10^{-3} and to 11.9 at the higher learning rates. The auxiliary objective is shaping the representation.

The distinction is therefore between structural repair and optimizer conditioning. Volume control can repair at-site failures while failing to control the global training dynamics. This is the central split of the paper: collapse prevention transfers locally, but EM conditioning does not automatically propagate through the supervised corridor.

8.4 Loss-Feature Decoupling Partially Replicates

The high-learning-rate Adam runs recover one part of the earlier single-layer phenomenology: lower auxiliary loss need not mean better task features. From Adam 10^{-3} to 10^{-1}, the regularization loss decreases from -57.3 to -265.9 and minimum variance explodes from 7.80 to 30628.4. Accuracy, however, barely moves: 96.38%, 96.37%, and 96.58%.

This is not a useful improvement in the representation. It shows that the volume-control objective has degrees of freedom that are largely orthogonal to classification accuracy once the useful level of spread and decorrelation has been reached. Adam at 10^{-3} is therefore the sensible operating point for the rest of the supervised experiments: it achieves high accuracy, low redundancy, and finite variance without driving the auxiliary geometry to an irrelevant extreme.

8.5 What Experiment 3 Does and Does Not Establish

Experiment 3 establishes the observational boundary. The same local objective that repairs collapse in Results I does not preserve the single-site optimizer signature under ordinary supervised training. This is already enough to reject the naive inheritance claim that a network with an EM site should inherit EM conditioning throughout the composed model.

It does not, by itself, prove the mechanism. The optimizer sweep is consistent with the theory that the CE path through W_2 contributes a larger, ill-conditioned foreign gradient at W_1, but Table 5 does not measure that gradient directly. Results III therefore performs the causal test: it measures CE/EM gradient dominance, measures the curvature dichotomy, and cuts the CE-to-W_1 corridor with stop-gradient. If the conditioning signature returns when the corridor is cut or when the EM path dominates, then the failure in Experiment 3 is not a failure of volume control. It is the locality boundary of implicit EM.

9 Results III: The Mechanism and Causal Test (Experiment 4)

Experiment 3 established the boundary observationally; Experiment 4 asks why. It tests three mechanism predictions—CE-path dominance at small λ, the curvature dichotomy, and causal restoration under stop-gradient—summarized against outcomes in the ledger below.

9.1 Predictions Ledger

Table 6 summarizes the predictions already tested in the supervised study. Because the claims are causal, the ledger separates predictions, measurements, and the one interpretation corrected after measurement.

Test	Prediction	Measured	Verdict
Exp1	Supervision does not give intermediate volume control	Baseline has min variance zero and ≥ 1 dead unit in every seed	✓
Exp1	Partial volume control is worse than none	Redundancy increases from 20.5 to 187.0; accuracy is worst	✓
Exp3	Optimizer signature disappears under composition	SGD spans 83.8–96.2%; Adam helps	✓
Exp4-P1	CE dominates EM at small λ	Late CE/EM ratio is 67–95× at $\lambda = 0.001$	✓
Exp4-P2	LSE curvature stays below 1/2; CE-path curvature is larger	LSE bound holds; CE path is 13–100× larger	✓
Exp4-P3	Stop-gradient restores conditioning	Learning-rate spread falls from 10.9 to 1.6 points	✓
Exp4	$\lambda = 1$ joint behaves like stop-gradient (dominance, not connectivity)	Spread 1.2 vs 1.6 points; both are nearly flat	✓
Exp4	Restored conditioning costs task alignment	EM-dominated features probe $\sim 88\%$ vs $\sim 96\%$	✓

Table 6: Prediction ledger for the supervised study. The table separates mechanism predictions, measurements, and post-measurement interpretation. One interpretation was corrected after measurement: the early heuristic claim that $\lambda = 0.001$ implied a 1000× gradient ratio was wrong; the measured late-training ratio is 67–95× (Section 9, gradient competition). The full trajectory is wider because of early transients. The mechanism is unchanged; the magnitude was corrected.

9.2 Experimental Arms

The causal experiment uses the model and auxiliary objective from Section 6 with $K = 25$. Four arms differ only in how the supervised and local objectives reach W_1:

The conditioning sweep trains each arm with SGD over a 1000× learning-rate range, $\eta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}\}$, using 3 seeds. Feature quality is measured by a fixed-protocol linear probe trained on frozen intermediate distances. This separates representation quality from the training success of the supervised head at a particular learning rate.

Arm	Objective	CE reaches W_1?
Joint, $\lambda = 0.001$	$L_{\text{CE}} + 0.001 L_{\text{aux}}$	yes
Joint, $\lambda = 0.03$	$L_{\text{CE}} + 0.03 L_{\text{aux}}$	yes
Joint, $\lambda = 1$	$L_{\text{CE}} + L_{\text{aux}}$	yes
Stop-gradient	$L_{\text{CE}}(\text{head}(\text{stopgrad}(y))) + L_{\text{aux}}$	no

Table 7: Arms for the causal experiment. The stop-gradient arm leaves the supervised head, labels, data, and architecture in place, but blocks the CE gradient from updating W_1.

9.3 Cutting the Corridor Restores Learning-Rate Insensitivity

The stop-gradient arm restores the most visible optimizer signature from the single-layer implicit-EM study [12]: feature quality becomes nearly insensitive to learning rate. Table 8 shows linear-probe accuracy across the 1000× SGD sweep.

Arm	10^{-4}	10^{-3}	10^{-2}	10^{-1}	Spread
Joint, $\lambda = 0.001$	85.30 ± 2.53	92.99 ± 0.15	95.05 ± 0.06	96.19 ± 0.17	10.9
Joint, $\lambda = 0.03$	86.42 ± 0.28	90.53 ± 0.04	91.78 ± 0.13	92.23 ± 0.29	5.8
Joint, $\lambda = 1$	88.35 ± 0.07	88.20 ± 0.10	88.41 ± 0.09	87.26 ± 0.52	1.2
Stop-gradient	88.37 ± 0.08	88.10 ± 0.07	87.94 ± 0.16	86.73 ± 0.50	1.6

Table 8: Linear-probe accuracy on frozen intermediate distances. Values are mean ± standard deviation over 3 seeds, in percent.

With ordinary joint training at $\lambda = 0.001$, probe accuracy varies by 10.9 points across the learning-rate range. When the CE gradient is prevented from reaching W_1, the spread falls to 1.6 points. The network is still supervised: the same labels, CE loss, architecture, and head training remain. Only the corridor gradient is removed. Supervision itself therefore does not destroy the conditioning signature; the pullback through the corridor does.

The $\lambda = 1$ joint arm is equally important. It is fully connected: the CE gradient still reaches W_1. Yet it is indistinguishable from stop-gradient on the conditioning metric. This shows that connectivity is not the decisive variable. The corridor manufactures a foreign, non-EM contribution, but that contribution controls the effective landscape only when it dominates the local EM contribution.

The intermediate value $\lambda = 0.03$ lies between the two regimes. Its spread is 5.8 points, roughly halfway between the CE-dominated and EM-dominated arms. This is the first evidence for the graded claim: the transition is not a phase change at the presence or absence of a graph edge. It follows dominance.

9.4 The EM Site Stabilizes Only When the EM Path Is Live

We also track the epoch at which test L_{LSE} reaches 95% of its final value. This is a stability metric for the local EM site, not a general classifier-training metric. In the CE-dominated arm, the classifier can train successfully while the local LSE objective is mostly along for the ride.

Under stop-gradient, the local site stabilizes in a narrow band, roughly 22–32 epochs across a 1000× learning-rate range. Under CE-dominated joint training, the same metric is erratic and sometimes reaches the training budget. This supports the same conclusion as the probe sweep:

Arm	10^{-4}	10^{-3}	10^{-2}	10^{-1}
Stop-gradient	32.0	27.7	21.7	23.7
Joint, $\lambda = 0.001$	32.3	15.0	40.0	37.7

Table 9: Epoch when test L_{LSE} reaches 95% of its final value. The two extreme arms (CE-blocked local EM path vs. strongly CE-dominated) are shown because they bound the contrast; this metric was not recorded for the intermediate $\lambda = 0.03$ and $\lambda = 1$ arms. The CE-dominated arm should be read as an EM-site stability contrast, not as a classifier failure metric.

when W_1 is trained by the local EM objective, the site has the fixed-time behavior of the single-layer model; when the CE path dominates, the local EM site does not stabilize as an autonomous mixture.

9.5 Gradient Dominance, Not Gradient Opposition

The mechanism is visible directly at the distance interface. In the reported diagnostic, the CE and auxiliary gradients with respect to d are measured on the same batch during Adam training at learning rate 10^{-3}. The conditioning sweep uses SGD, so the table should be read as a site-level gradient diagnostic rather than a matched-optimizer measurement. Table 10 reports the ratio

$$\frac{\|\nabla_d L_{\text{CE}}\|}{\lambda \|\nabla_d L_{\text{aux}}\|}$$

and the cosine between the two gradient directions.

Arm	Early CE/EM ratio	Late CE/EM ratio	Cosine
Joint, $\lambda = 0.001$	0.18–0.74×	67–95×	−0.05 to 0.05
Joint, $\lambda = 1$	0.00018–0.00074×	0.018–0.025×	−0.07 to 0.08

Table 10: Gradient competition at the intermediate distance coordinates. The ratio uses the weighted auxiliary gradient, $\lambda \|\nabla_d L_{\text{aux}}\|$. Early values are the first recorded measurement for each seed; late values are the final recorded measurement.

This measurement corrects an earlier heuristic interpretation. Since $\lambda = 0.001$, one might expect a 1000× ratio. The measured ratio is 67–95× late in training, because the raw auxiliary gradient is larger than the raw CE gradient. The full trajectory is wider, with early transients before the CE path takes over. The conclusion is unchanged: CE dominates at the operating point, but the causal variable is the measured gradient ratio, not λ itself.

The cosine measurement shows no detectable alignment between the two paths. In $K = 25$ dimensions, random directions have cosine standard deviation approximately $1/\sqrt{25} = 0.2$, so a cosine near zero should not be oversold as special high-dimensional orthogonality. The conservative statement is enough: the EM and CE gradients are not detectably aligned or opposed. The EM signal is drowned by a larger foreign signal, not cancelled by an antagonistic one.

The prediction is that the dominance ratio is a property of the loss geometry at the site, not an Adam-specific artifact; a matched-SGD gradient measurement is a remaining strengthening check.

9.6 Curvature: The LSE Bound Holds, the CE Path Is Large

Proposition 3 says that composition forfeits the uniform LSE bound; it does not prove that the composed curvature is necessarily large. The curvature measurements answer that empirical question. Power iteration on exact Hessian-vector products estimates the spectral norm of the Hessian with respect to the intermediate distances d for both the local LSE term and the CE path through NLS and the head.

The local LSE curvature obeys the theory at every measured point: it stays below 1/2 in every arm and seed. In the stop-gradient arm, it converges to 0.4992–0.4997, essentially saturating the bound. The bound is achieved when responsibility mass concentrates equally on two components. Under protected local-EM training, the mixture sharpens until cluster boundaries become two-way ties, and the Hessian detects exactly this geometry.

By contrast, the CE-path curvature in a representative joint $\lambda = 0.001$ trajectory wanders over values such as 6.5, 17.2, and 7.8 across training. At the same coordinates, this is roughly 13–100$\times$ larger than the LSE curvature, non-monotone, and consistent with dependence on the learned spectrum of W_2. This is the empirical half of the theory: composition removes the parameter-independent guarantee, and in the trained network the resulting curvature is actually large and changing.

The 1/2 bound is therefore more than a worst-case constant. Distance from 1/2 acts as a site-level diagnostic of mixture sharpness. Under EM-dominated training it saturates; under CE-dominated training, the LSE curvature stagnates well below the bound, ending between 0.158 and 0.236 across seeds. This indicates diffuse responsibilities and an under-formed local mixture.

9.7 Conditioning Is Restored at a Price

The EM-dominated arms are better conditioned but less task-aligned. Their frozen features probe at about 88%, while the CE-dominated arm reaches about 96%. This is not surprising in the stop-gradient arm: W_1 does not receive label information. The point is not that unsupervised local features beat supervised features. They do not.

The point is that conditioning is a real property with a real cost. The EM regime buys learning-rate robustness, seed-to-seed reproducibility, stable local convergence, and zero dead units; it does not buy output accuracy for free. This is why the λ-resolution curve is central: the trade-off should be shown as a continuum along the measured CE/EM gradient ratio, not as a binary comparison between supervised and unsupervised training.

10 Discussion

The results draw a boundary around the implicit-EM interpretation of neural training. The gradient-responsibility identity is real, and it has observable consequences inside a supervised network. But those consequences are local. A network can contain an EM site, benefit from local volume control at that site, and still fail to inherit the optimizer conditioning of an isolated EM model.

10.1 Local, Graded, and Structural

The three adjectives in the main claim refer to different parts of the evidence.

Implicit EM is *local*: the responsibility structure exists at an LSE or softmax site and extends to parameters only when the path preserves component identity—it does not automatically penetrate a learned dense map. The theory is the formal version of this. When the responsibility vector updates private component parameters directly, Proposition 1 gives a parameter-independent curvature bound; when it is pulled through a learned dense map, Proposition 3 shows the same bound no longer controls the composed path. Sections 8 and 9 show the empirical consequence: ordinary supervised composition restores learning-rate sensitivity and an Adam advantage.

Implicit EM is *graded*. The decisive variable is not whether the computation graph contains a path from CE to W_1. The $\lambda = 1$ joint arm keeps that path intact and nevertheless behaves like the stop-gradient arm on the conditioning metric. Conversely, the $\lambda = 0.001$ joint arm is CE-dominated and behaves like an ordinary supervised network. The reported CE/EM gradient-ratio diagnostic identifies the causal variable as dominance, not graph connectivity alone.

Implicit EM is *structural*. Volume control repairs failures at a local EM interface: dead units, collapsed variance, and redundancy. Dense learned composition creates a different failure: the foreign CE contribution reaching W_1 is not a private per-component responsibility update. The two failure classes require different repairs. At-site degeneracy is repaired by local volume control; structural loss of conditioning is repaired only by isolation, dominance, or compatible transport.

10.2 What Transfers

The part that transfers is the local mixture geometry. As Results I established, supervised learning does not by itself supply volume control to an intermediate site, and only full volume control (not competition alone, and not the anti-collapse term alone) eliminates dead units and redundancy across widths. The local EM site is not decorative and the auxiliary objective is not inert: it moves the structural health of the representation in the direction the decoder-free encoder study predicted.

The Var+Decorr control—which also performs well—is useful rather than damaging. It separates ordinary regularization from the narrower EM-specific claim: volume control is good regularization on its own, and when attached to an NLS site it additionally stabilizes the calibrated distance-to-responsibility geometry.

10.3 What Does Not Transfer

The single-site optimizer signature does not transfer through ordinary dense supervised composition. In the isolated implicit-EM model, SGD was unusually learning-rate insensitive and Adam had little to correct. In the supervised network, SGD spans a wide accuracy range across learning rates and Adam becomes useful again. The local objective remains present, but it no longer controls the training dynamics at W_1 when the supervised path dominates.

The causal experiment explains why. Cutting the CE-to-W_1 corridor restores the learning-rate-insensitive signature. Increasing λ until the EM path dominates restores the same signature even without cutting the graph. The failure is therefore not supervision per se and not the mere existence of a second layer. It is the dominance of a composed, non-EM gradient at the parameters whose conditioning we hoped the local EM site would control.

10.4 Dominance Resolves the Apparent Tension

There is an apparent tension between two statements: dense backpropagation is parameter sharing in disguise, but the fully connected $\lambda = 1$ arm behaves like stop-gradient. The resolution is to separate structure from dominance.

The structural statement explains the form of the foreign contribution. Once CE is pulled back through W_2, the gradient at W_1 is not a posterior over private hidden components, and its curvature can depend on the learned spectrum of the corridor. This is the neural version of a non-separable M-step.

The dominance statement explains whether that contribution matters. The update at W_1 is a sum:

$$\nabla_{W_1} L_{\text{total}} = \nabla_{W_1} L_{\text{CE}} + \lambda \nabla_{W_1} L_{\text{aux}}.$$

When the first term dominates, the effective landscape is the composed supervised landscape. When the second term dominates, the local EM geometry sets the behavior even though the CE path still exists. A denser λ-resolution curve is therefore the natural next strengthening analysis: the causal variable is the measured gradient ratio, not the nominal value of λ and not the binary presence of an edge.

10.5 Generative Versus Discriminative Is the Phenomenon, Not the Mechanism

One deflationary reading is that the paper only rediscovers a familiar generative-versus-discriminative distinction. The experiments are compatible with that high-level framing, but they are more specific. The same architecture is trained with the same data and the same local site while the route by which CE reaches W_1 is changed. The mechanistic measurements are specific: the LSE curvature obeys the $1/2$ bound, the CE-path curvature is larger and changing, the reported gradient diagnostic shows CE dominance at small λ, and removing that path restores the conditioning signature.

The result is therefore not just that two objectives differ. It identifies how the discriminative objective reaches into a local EM site and when that gradient controls the update. This is the useful level of explanation for model design: it tells us which path must be blocked, weakened, or made compatible if we want local EM conditioning to survive.

10.6 The Accuracy Trade-Off

The EM-dominated features probe around 88%, while the CE-dominated features probe around 96%. That gap should be stated plainly. Well-conditioned optimization of a less task-aligned local objective does not beat direct supervised optimization of the task objective.

This does not make the conditioning result unimportant. It clarifies what the property is for. EM conditioning buys robustness to learning-rate choice, lower seed-to-seed variability, stable local convergence, and reliable use of all components. Those benefits matter most when the local objective is not being asked to replace the supervised head at the output: for layer-local representation learning, protected intermediate objectives, or architectures where local mixture structure is itself the desired object. In an ordinary end-to-end classifier, the task head can trade away that structure for accuracy.

10.7 Pretraining and Fine-Tuning

The results suggest a concrete interpretation of discriminative fine-tuning. If a representation is first shaped by a local EM objective and then exposed to a dominant supervised corridor, fine-tuning may not simply refine the EM solution. It may move the model into a different regime whose features are more task-aligned but less locally EM-conditioned.

The current evidence is suggestive rather than complete. The CE and EM paths show no detectable alignment, and the EM-dominated and CE-dominated arms occupy different accuracy and conditioning regimes. A basin or linear-mode-connectivity analysis, with hidden units aligned before interpolation, would test the stronger claim: whether the two regimes are connected by a low-loss path or separated by a barrier after accounting for permutation symmetry.

10.8 Design Implications

The practical rule is simple: create EM sites locally, and do not expect ordinary dense backpropagation to preserve their conditioning. If the goal is local mixture structure, the objective should reach the relevant parameters directly. If a supervised or foreign objective must share those parameters, its gradient must either be blocked, balanced so the EM path dominates when desired, or transported through operations that preserve the simplex structure.

This rule also explains why stop-gradient is not merely a training trick in this setting. It is a structural intervention: it turns a shared, non-separable update back into a local one. Conversely, increasing λ is not just "more regularization." It changes the effective optimization regime by changing which gradient controls W_1.

The stochastic-map case is the natural remaining taxonomy cell. A nonnegative row-normalized corridor can preserve nonnegativity and total mass at first order, unlike an arbitrary dense signed matrix. This variant should be interpreted as a test of partial preservation, not as a load-bearing headline result. If row normalization damages task accuracy enough to confound the probe, the result belongs as a caveated appendix experiment.

10.9 Relation to Attention

The companion attention work asks whether EM-like structure propagates through transformer depth. The locality result here gives the controlled answer that such propagation should not be assumed. A standard transformer has an output CE site that owns an objective; intermediate attention softmaxes are computation inside a larger objective, not automatically owned EM sites. Their gradients are shaped by foreign downstream losses and learned linear maps.

The connection should remain narrow in this paper. The supervised study is the controlled proof of the locality principle. Transformer experiments are the "in the wild" test of the same design rule. The stochastic-map variant is the bridge: attention transports values through a stochastic map, which may preserve some responsibility-like structure while still failing to preserve full optimizer conditioning through the surrounding learned maps.

11 Limitations and Threats to Validity

The paper makes a narrow claim about where implicit-EM conditioning is preserved and where it is lost. The evidence is intentionally controlled rather than broad. This section separates threats to the internal interpretation of the experiments from limitations on external scope.

11.1 Threats to Internal Validity

Feature quality is measured primarily by a linear probe. The main accuracy trade-off compares frozen intermediate representations with a fixed-protocol linear probe. This is the right first probe because it measures linearly accessible class information without allowing the downstream evaluator to repair the representation. It is not the only possible measure of feature quality. A kNN probe and a small MLP probe on the same frozen distances would test whether the 88% versus 96% ordering is a property of the representation or a property of the probe. If the ordering survives these checks, the interpretation strengthens: EM-dominated features are genuinely less task-aligned, not merely less linearly separable.

The curvature measurements depend on numerical estimation. The curvature claims use spectral norms estimated by power iteration on fixed batches. The qualitative contrast is large: the local LSE curvature remains below 1/2, while the composed CE-path curvature is much larger and non-monotone. Still, the exact reported ratios can depend on batch identity, batch size, and the number of power-iteration steps. The final presentation should report sensitivity to these choices. The key internal check is that the LSE bound remains stable and the CE-path curvature remains separated from it under reasonable measurement variations.

The 1/2-saturation diagnostic is demonstrated at one site. The paper uses proximity to the LSE curvature bound as a diagnostic of mixture sharpness. In this study, the diagnostic behaves as predicted: EM-dominated training approaches the 1/2 bound, while CE-dominated training leaves the local site diffuse. That supports the diagnostic locally, but not its general use across architectures. Claims about 1/2-saturation should therefore remain modest until the same measurement is repeated across depths, datasets, or different EM-site parameterizations.

Stop-gradient is a clean but strong intervention. The stop-gradient arm blocks the CE gradient from reaching W_1 while keeping the data, labels, architecture, CE loss, and supervised head in place. This makes it a strong causal test of the corridor-gradient hypothesis. It is also a large intervention: it changes the optimization problem for W_1 from joint supervised training to protected local training. The interpretation rests on the fact that the intervention changes exactly the path under dispute. It does not show that stop-gradient is the best practical method; it shows that the CE corridor is sufficient to destroy the conditioning signature when it dominates.

Gradient dominance should be measured under the matched optimizer. The gradient-competition measurements are taken during Adam training at learning rate 10^{-3}, while the conditioning sweep in the causal experiment uses SGD. The dominance ratio is expected to reflect the loss geometry at the distance interface rather than an Adam-specific artifact, but the cleanest version of the result would measure the same CE/EM ratio under the same optimizer and learning-rate sweep used for the conditioning result. Until then, the optimizer mismatch should be stated explicitly.

The optimizer-signature restoration is incomplete. The stop-gradient arm restores SGD learning-rate insensitivity at W_1, and the ordinary composed arm shows a renewed advantage for Adam. The complementary Adam stop-gradient arm has not yet been reported. That run would test the remaining half of the optimizer-signature claim: whether Adam loses its advantage when the local EM objective owns the W_1 update.

Cosine near zero is a weak alignment claim. The measured CE and EM gradients have cosine near zero. In $K = 25$ dimensions, random directions also have cosine standard deviation about $1/\sqrt{25} = 0.2$. Therefore the correct conclusion is not strong geometric orthogonality. The conservative conclusion is that there is no detectable alignment or opposition at the measured scale. This still supports the mechanism: the EM signal is drowned by a larger foreign signal, not cancelled by a directly opposing one.

The graded claim is currently supported by few points. The causal experiment includes $\lambda = 0.001$, $\lambda = 0.03$, $\lambda = 1$, and stop-gradient. This is enough to show that the effect is not binary: $\lambda = 0.03$ lies between the CE-dominated and EM-dominated regimes, and $\lambda = 1$ behaves like stop-gradient despite full connectivity. A denser λ-resolution curve, plotted against the measured CE/EM gradient ratio, would make the graded claim much stronger and is the natural next analysis for this result.

The learning-rate sweeps mainly show low-rate slowness. The reported sweeps show large final-accuracy spreads across learning rates, but the SGD arms mostly improve monotonically over the tested range. Thus the current spread metric primarily captures undertraining at small learning rates, not an observed upper instability edge. A larger learning-rate rung or time-to-threshold analysis would make the smoothness interpretation sharper.

Some structural metrics are close to their training losses. Dead-unit and redundancy measurements are related to the variance and decorrelation penalties used in the auxiliary objective. This does not make the repair trivial: the cross-metrics matter, for example the variance-only arm removes dead units while making redundancy much worse. Still, the overlap between objective terms and reported health metrics should be read as a targeted repair test rather than as an independent representation-quality measure.

The auxiliary objective has escape directions. The clean curvature propositions apply to the LSE contribution at the local responsibility site. The full auxiliary objective also includes variance and decorrelation terms, and high-learning-rate Adam can drive the variance term to very large values without improving task accuracy. The reported operating point is chosen below this runaway regime; the theory should not be read as a global smoothness proof for the full auxiliary objective.

The basin interpretation remains incomplete. The discussion interprets discriminative fine-tuning as a dominant foreign gradient that can move the model away from local EM structure. The present evidence supports this as a mechanism-level interpretation, but it does not yet prove that EM-dominated and CE-dominated solutions occupy different basins. A linear-mode-connectivity analysis with hidden-unit alignment is needed before making a stronger basin claim. Without alignment, apparent barriers could be artifacts of component permutation.

11.2 Limitations of Scope

Dataset and scale. The experiments use MNIST and small supervised networks. This is appropriate for isolating the mechanism, but it limits external validity. The paper should not claim state-of-the-art performance, broad deep-learning optimization coverage, or automatic transfer to modern large-scale architectures. The claim is that the controlled setting exposes a locality boundary that larger systems must respect when the same structural conditions are present.

Single main architecture. The experimental architecture has one intermediate EM site followed by a dense supervised head. This is the architecture needed to test the corridor hypothesis directly. Other architectures may include normalization, residual paths, attention-style stochastic maps, convolutional sharing, or multiple local sites. The theory predicts which operations should preserve or destroy the simplex structure, but those variants require separate experiments.

The formal results are sufficient conditions, not an if-and-only-if. The private-parameter propositions show that local LSE sites can inherit a parameter-independent curvature bound. The composition proposition shows that a learned dense map forfeits that guarantee. These are not universal necessity claims. Some composed systems may remain well conditioned for unrelated reasons, and some private systems may fail because of effects outside the clean theory, such as hard gates or bad data geometry.

The theory does not prove composed curvature is always large. Proposition 3 is a guarantee-forfeiture statement. It shows that the clean 1/2 LSE bound no longer controls the composed CE path because the bound can depend on the learned spectrum of W_2. The empirical measurements show that the composed curvature is large in the studied network. The theorem alone does not imply that every dense composition has large curvature at every point.

LayerNorm is handled empirically but not fully in the clean proof. The measured models include LayerNorm where specified, so the reported conditioning and curvature results include its effect. The clean propositions omit LayerNorm to isolate the LSE and dense-corridor geometry. A fully expanded proof would insert the LayerNorm Jacobian and additional input-dependent terms. This would complicate the constants but would not restore the parameter-independent LSE guarantee through a learned dense corridor.

No convergence theorem is provided for the composed system. The paper analyzes local curvature and measures optimization signatures, but it does not prove convergence of the full supervised system. The results should be read as a conditioning and mechanism study, not as a global optimization theory for neural networks.

Future strengthening analyses. A denser λ-resolution curve, plotted against the measured CE/EM gradient ratio, would strengthen the graded claim. A basin or linear-mode-connectivity analysis with hidden-unit alignment would test whether EM-dominated and CE-dominated checkpoints occupy distinct basins. A nonnegative row-normalized corridor would test the simplex-compatible preservation class separately from the dense-corridor arms. These analyses are useful extensions, but they are not load-bearing for the current volume-control, stop-gradient, dominance, and curvature results.

The stochastic-map bridge is not load-bearing. A row-normalized nonnegative corridor is an important preservation-class test and a useful bridge to attention, but it should not carry the main claims. If the stochastic-map arm is clean, it strengthens the operation taxonomy. If it is confounded by reduced task accuracy or head capacity, it should remain an appendix-style result. The core claims are already tested by the volume-control ablations, optimizer sweep, stop-gradient intervention, gradient dominance, and curvature measurements.

12 Conclusion

Log-sum-exp objectives make responsibilities appear as gradients. This gives standard neural objectives a precise implicit-EM interpretation, but the interpretation has a boundary. It is a statement about the coordinates seen by an LSE or softmax objective, not a guarantee that a whole composed network inherits EM's optimizer geometry.

The theory identifies that boundary. In distance coordinates, the local LSE Hessian is uniformly bounded by 1/2. When component parameters are private and affine, this yields a parameter-independent curvature bound and explains the single-site optimizer signature observed in earlier work. When the same site is trained through a learned dense corridor, the leading term of the composed CE Hessian is a conjugated softmax Hessian whose scale depends on the learned map. The local guarantee is therefore forfeited.

The supervised experiments bear out the split in this controlled setting: volume control transfers to an intermediate EM site, but the single-site optimizer conditioning does not survive ordinary composition—and is restored, causally, by cutting the corridor or letting the EM path dominate.

The resulting rule is local, graded, and structural. EM conditioning survives when responsibility gradients reach the relevant parameters directly or when the EM path dominates the update; ordinary dense backpropagation should not be expected to preserve it. Simplex-compatible transport remains a theoretical first-order preservation class rather than a load-bearing empirical claim here. Two consequences point beyond this paper. For architecture, the result motivates building depth as protected local EM sites rather than expecting a single owned objective to condition every layer. For interpretation, it says the intermediate softmax sites of standard attention architectures, owning no objective of their own, should not be read as automatically inheriting EM's optimization structure. Knowing which properties transfer, which fail, and which interventions make them survive is the boundary the implicit-EM framework now provides.

References

[1] Eugene Belilovsky, Michael Eickenberg, and Edouard Oyallon. Greedy layerwise learning can scale to imagenet. In *International Conference on Machine Learning*, 2019.

[2] Dankmar Bohning. Multinomial logistic regression algorithm. *Annals of the Institute of Statistical Mathematics*, 1992.

[3] Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*, 2018.

[4] Arthur P. Dempster, Nan M. Laird, and Donald B. Rubin. Maximum likelihood from incomplete data via the em algorithm. *Journal of the Royal Statistical Society: Series B*, 1977.

[5] Rahim Entezari, Hanie Sedghi, Olga Saukh, and Behnam Neyshabur. The role of permutation invariance in linear mode connectivity of neural networks. In *International Conference on Learning Representations*, 2022.

[6] Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M. Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020.

[7] Geoffrey E. Hinton. The forward-forward algorithm: Some preliminary investigations, 2022. arXiv:2212.13345.

[8] Geoffrey E. Hinton, Simon Osindero, and Yee-Whye Teh. A fast learning algorithm for deep belief nets. *Neural Computation*, 2006.

[9] Max Jaderberg, Wojciech Marian Czarnecki, Simon Osindero, Oriol Vinyals, Alex Graves, David Silver, and Koray Kavukcuoglu. Decoupled neural interfaces using synthetic gradients. In *International Conference on Machine Learning*, 2017.

[10] Radford M. Neal and Geoffrey E. Hinton. A view of the em algorithm that justifies incremental, sparse, and other variants. *Learning in Graphical Models*, 1998.

[11] Andrew Y. Ng and Michael I. Jordan. On discriminative vs. generative classifiers: A comparison of logistic regression and naive bayes. *Neural Information Processing Systems*, 2002.

[12] Alan Oursland. Gradient descent as implicit em in distance-based neural models, 2025. arXiv:2512.24780.

[13] Alan Oursland. Deriving decoder-free sparse autoencoders from first principles, 2026. arXiv:2601.06478.

[14] Mohammad Pezeshki, S. O. Kaba, Yoshua Bengio, Aaron Courville, Doina Precup, and Guillaume Lajoie. Gradient starvation: A learning proclivity in neural networks. In *Neural Information Processing Systems*, 2021.

[15] Lei Xu and Michael I. Jordan. On convergence properties of the em algorithm for gaussian mixtures. In *Neural Information Processing Systems*, 1996.

[16] Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Neural Information Processing Systems*, 2020.